10 In this question you must show detailed reasoning.

Fig. C2.2 indicates that the curve $y = \dfrac{4x(\pi - x)}{\pi^2} - \sin x$ has a stationary point near $x = 3$.

- Verify that the x-coordinate of this stationary point is between 2.6 and 2.7.

- Show that this stationary point is a maximum turning point. **[5]**